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In [1]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Sample data
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
Y = np.array([35, 50, 65, 70, 85])

# Create model
model = LinearRegression()

# Train model
model.fit(X, Y)

# Prediction
x_input = float(input("Enter hours studied: "))

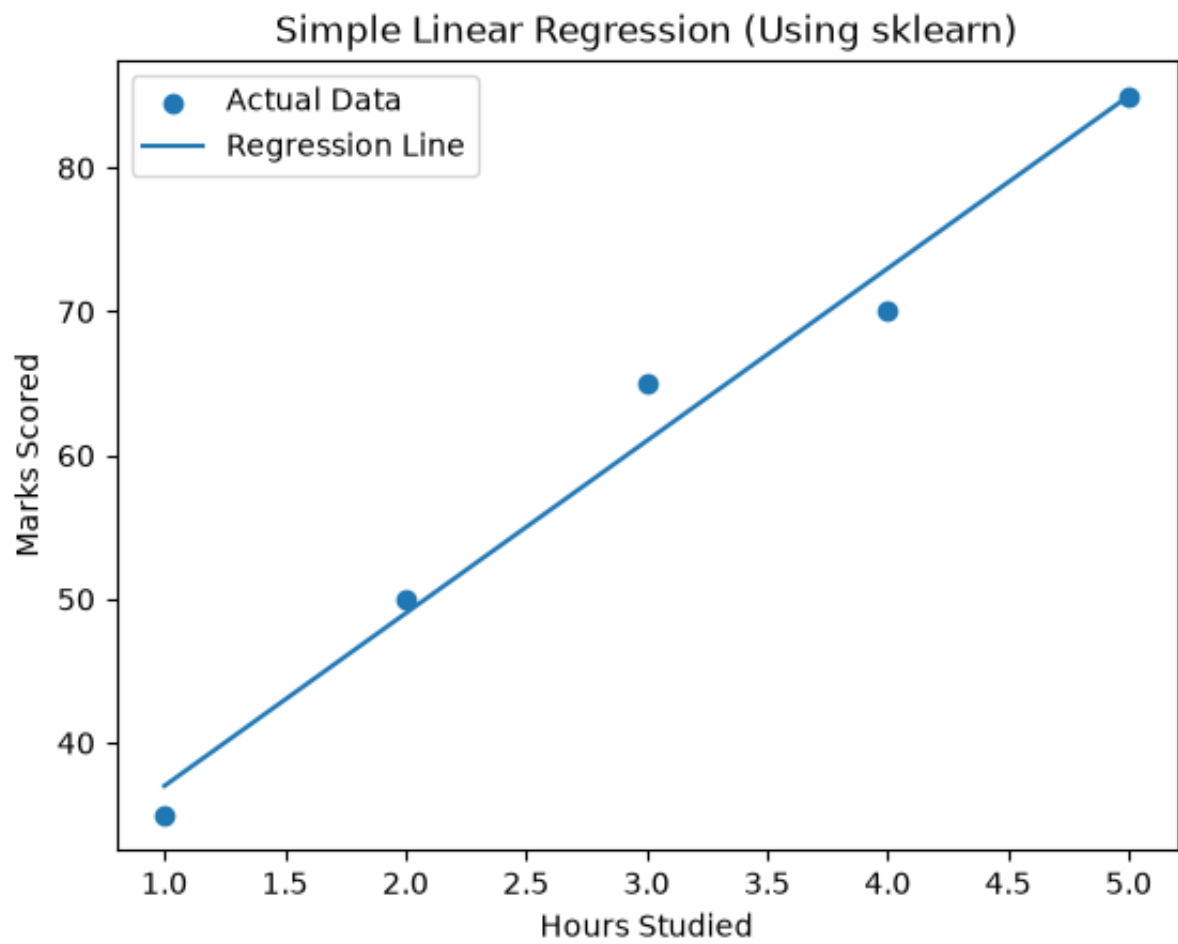
predicted_marks = model.predict([[x_input]])

print("Predicted Marks:", predicted_marks[0])

# Plot
Y_pred = model.predict(X)

plt.scatter(X, Y, label="Actual Data")
plt.plot(X, Y_pred, label="Regression Line")
plt.xlabel("Hours Studied")
plt.ylabel("Marks Scored")
plt.title("Simple Linear Regression (Using sklearn)")
plt.legend()
plt.show()
```

Predicted Marks: 97.0



In []: